Experiment 3: Download and Install Hadoop Framework (Windows)

Aim

To download, install, configure environment paths, and execute startup scripts for a single-node

Apache Hadoop cluster on a Windows operating system. Procedure
1. Download a stable binary release archive of Apache Hadoop from the official Apache website and

extract it directly into a clean root drive folder like C:\hadoop. 2. Locate and download compatible Windows native binary dependencies (winutils.exe and
hadoop.dll) for your specific Hadoop version, and copy them into the C:\hadoop\bin directory. 3. Access the Windows System Environment Variables panel and create new system variables
named HADOOP_HOME pointing to C:\hadoop and JAVA_HOME pointing to your JDK installation path. 4. Edit the Windows system Path variable to append %HADOOP_HOME%\bin and %HADOOP_HOME%\sbin to
allow cluster commands to execute globally from any Command Prompt. 5. Modify the Hadoop configuration scripts inside C:\hadoop\etc\hadoop\, specifically editing core-

site.xml to specify the default HDFS URI (hdfs://localhost:9000) and hdfs-site.xml to set

replication factor properties to 1. 6. Open a fresh Windows Command Prompt as an Administrator and execute the HDFS storage
structural layout initialization command: hdfs namenode -format. 7. Initialize the standalone Windows pseudo-distributed cluster network operations by executing the

native startup batch driver script start-dfs.cmd followed by start-yarn.cmd. Commands

cmd
:: 1. Verify your system Java Environment status before installation
java -version

:: 2. Set environment paths via command line (Alternative to GUI Environment Panel)
setx /M HADOOP_HOME "C:\hadoop"
setx /M JAVA_HOME "C:\Program Files\Java\jdk-11"
setx /M PATH "%PATH%;%HADOOP_HOME%\bin;%HADOOP_HOME%\sbin"

:: 3. Navigate to configuration directory to review setups
cd C:\hadoop\etc\hadoop

:: 4. Format the master HDFS filesystem NameNode structure

hdfs namenode -format

:: 5. Launch HDFS Storage System Services
cd C:\hadoop\sbin
start-dfs.cmd

:: 6. Launch YARN Resource Management Services
start-yarn.cmd

:: 7. Check active background Java processes running on Windows jps

Use code with caution. Execution & Output

text
C:\hadoop\sbin>hdfs namenode -format
26/06/03 11:15:22 INFO namenode.NameNode: STARTUP_MSG: /**
STARTUP_MSG: Formatting NameNode storage directory successfully... **/

C:\hadoop\sbin>start-dfs.cmd
:: [A new Command Prompt window automatically spawns launching the NameNode process] :: [A second Command Prompt window automatically spawns launching the DataNode process]

C:\hadoop\sbin>jps
14320 NameNode
15812 DataNode
11404 ResourceManager
16240 NodeManager
12908 Jps